1. The point through which the resultant gravitational weight of a physical body acts is termed its:

(A) Centre of Gravity (CG)
(B) Centroid
(C) Centre of Pressure
(D) Instantaneous Centre

2. The geometric center of a 2D plane area or lamina depending solely on its geometric boundary is the:

(A) Centroid
(B) Centre of Gravity
(C) Metacentre
(D) Centre of Percussion

3. For a homogeneous material in a uniform gravitational field, the Centre of Gravity, Centre of Mass, and Centroid:

(A) Coincide identically at the same point
(B) Are always at different locations
(C) Are undefined for composite shapes
(D) Depend on the temperature gradient

4. The first moment of area Q_y of a plane area A about the y-axis is mathematically defined as:

(A) Integral(x dA) = A * x_bar
(B) Integral(y dA) = A * y_bar
(C) Integral(x^2 dA)
(D) Integral(x*y dA)

5. The first moment of area about any centroidal axis passing through the centroid is always:

(A) Zero
(B) Equal to total area A
(C) Equal to infinity
(D) Equal to the moment of inertia

6. If a plane cross-section possesses an axis of symmetry, its centroid:

(A) Must lie somewhere on that axis of symmetry
(B) Must lie outside the section
(C) Cannot be determined by statics
(D) Is always at the origin (0,0)

7. If a cross-section possesses two orthogonal axes of symmetry (e.g. standard I-beam or circle), the centroid lies:

(A) At the geometric intersection of the two axes
(B) At the bottom boundary
(C) At the top flange apex
(D) Along the diagonal line

8. The First Theorem of Pappus-Guldinus states that the surface area generated by revolving a plane curve about a coplanar non-intersecting axis equals:

(A) Curve length times distance traversed by its centroid (2*pi*y_bar*L)
(B) Area times distance traversed by centroid
(C) Curve length squared times pi
(D) Two times area divided by length

9. The Second Theorem of Pappus-Guldinus states that the volume generated by revolving a plane area about a coplanar non-intersecting axis equals:

(A) Area times distance traversed by its centroid (2*pi*y_bar*A)
(B) Curve length times distance traversed
(C) Area squared times 2*pi
(D) Volume divided by centroid radius

10. In applying the theorems of Pappus-Guldinus, the axis of revolution must strictly:

(A) Not intersect the generating curve or area
(B) Pass directly through the centroid
(C) Be inclined at 45 degrees
(D) Be a curved spatial axis

11. The second moment of area of a planar figure about the x-axis (I_xx) is defined as:

(A) Integral(y^2 dA)
(B) Integral(x^2 dA)
(C) Integral(y dA)
(D) Integral(x*y dA)

12. In SI units, the second moment of area (area moment of inertia) is expressed in:

(A) m^4 or mm^4
(B) m^3 or mm^3
(C) N-m
(D) kg-m^2

13. Mass moment of inertia (resistance to angular acceleration) in SI units is expressed in:

(A) kg-m^2
(B) m^4
(C) N-m/s
(D) kg/m^2

14. The Polar Moment of Inertia J (or I_zz) for a 2D plane area lying in the x-y plane is defined as:

(A) Integral(r^2 dA) = I_xx + I_yy
(B) Integral(r dA)
(C) I_xx - I_yy
(D) sqrt(I_xx * I_yy)

15. The Perpendicular Axis Theorem states that for any flat planar lamina in the x-y plane:

(A) I_zz = I_xx + I_yy
(B) I_zz = I_xx - I_yy
(C) I_zz = sqrt(I_xx^2 + I_yy^2)
(D) I_xx = I_yy + I_zz

16. The Perpendicular Axis Theorem is strictly valid only for:

(A) Planar 2D laminas lying in a single plane
(B) Any general 3D spatial solid body
(C) Thin spherical shells only
(D) Hollow cylinders only

17. The Parallel Axis Theorem (Steiner's Theorem) for area moment of inertia is mathematically formulated as:

(A) I_axis = I_centroidal + A * d^2
(B) I_axis = I_centroidal - A * d^2
(C) I_centroidal = I_axis + A * d^2
(D) I_axis = I_centroidal + A * d

18. In the Parallel Axis Theorem I = I_G + A*d^2, the distance d represents:

(A) Perpendicular distance between centroidal axis and target axis
(B) Distance from origin to point
(C) Radius of gyration
(D) Length of the member

19. Among all parallel axes in a plane, the moment of inertia is minimum about:

(A) The axis passing through the centroid
(B) The base axis
(C) The top extreme fiber axis
(D) An axis at infinity

20. The Radius of Gyration k of a plane area A with moment of inertia I is defined by:

(A) k = sqrt(I / A)
(B) k = I / A
(C) k = sqrt(A / I)
(D) k = I^2 / A

21. The physical interpretation of radius of gyration k is:

(A) Distance at which entire area could be concentrated to give same MOI
(B) Distance to extreme fiber
(C) Centroidal height
(D) Diameter of gyration

22. For a rectangle of base b and height h, the centroidal moment of inertia I_xx is:

(A) b * h^3 / 12
(B) b * h^3 / 3
(C) b * h^3 / 36
(D) b^3 * h / 12

23. For a rectangle of base b and height h, the moment of inertia about its base edge is:

(A) b * h^3 / 3
(B) b * h^3 / 12
(C) b * h^3 / 4
(D) b * h^3 / 24

24. For a triangle of base b and height h, the centroidal moment of inertia I_xx is:

(A) b * h^3 / 36
(B) b * h^3 / 12
(C) b * h^3 / 3
(D) b * h^3 / 48

25. For a triangle of base b and height h, the moment of inertia about its base is:

(A) b * h^3 / 12
(B) b * h^3 / 36
(C) b * h^3 / 3
(D) b * h^3 / 4

26. The centroid of a right triangle of base b and height h measured from its right-angled corner is:

(A) (b/3, h/3)
(B) (b/2, h/2)
(C) (2b/3, 2h/3)
(D) (b/4, h/4)

27. For a solid circular area of radius R (diameter d), the centroidal moment of inertia I_xx is:

(A) pi * R^4 / 4 = pi * d^4 / 64
(B) pi * R^4 / 2
(C) pi * R^4 / 8
(D) pi * d^4 / 32

28. The polar moment of inertia J of a solid circular shaft of diameter d is:

(A) pi * d^4 / 32
(B) pi * d^4 / 64
(C) pi * d^4 / 16
(D) pi * d^4 / 4

29. The centroid of a semicircular area of radius R measured from its diametral base along the symmetry axis is:

(A) 4 * R / (3 * pi)
(B) 3 * R / (4 * pi)
(C) R / 2
(D) 2 * R / pi

30. The centroid of a semicircular thin wire arc of radius R from its base is:

(A) 2 * R / pi
(B) 4 * R / (3 * pi)
(C) R / 2
(D) R / pi

31. For a quarter-circle of radius R, the centroid coordinates (x_bar, y_bar) from the center are:

(A) (4R / 3pi, 4R / 3pi)
(B) (2R / pi, 2R / pi)
(C) (R/2, R/2)
(D) (R/3, R/3)

32. For a hollow circular pipe with outer diameter D and inner diameter d, the moment of inertia I_xx is:

(A) pi * (D^4 - d^4) / 64
(B) pi * (D - d)^4 / 64
(C) pi * (D^4 + d^4) / 64
(D) pi * (D^2 - d^2) / 4

33. The Product of Inertia I_xy of a plane area A is defined as:

(A) Integral(x * y dA)
(B) Integral(x^2 * y^2 dA)
(C) Integral(x dA) * Integral(y dA)
(D) I_xx * I_yy

34. If a cross-section possesses at least one axis of symmetry (e.g. x-axis or y-axis), its product of inertia I_xy is:

(A) Identically zero (I_xy = 0)
(B) Maximum
(C) Always negative
(D) Equal to J

35. Axes for which the product of inertia vanishes (I_xy = 0) are termed the:

(A) Principal Axes of Inertia
(B) Centroidal Conjugate Axes
(C) Secant Axes
(D) Eulerian Neutral Axes

36. The moments of inertia about the principal axes are called the:

(A) Principal Moments of Inertia (I_max and I_min)
(B) Polar Constants
(C) Gyration Radii
(D) Second Order Integrals

37. The sum of moments of inertia about any two mutually perpendicular axes through a point is:

(A) Constant (Invariant under coordinate rotation: I_xx + I_yy = J)
(B) Variable depending on angle
(C) Zero
(D) Equal to I_xy

38. Section Modulus Z for a beam cross-section subjected to bending is defined as:

(A) Z = I / y_max
(B) Z = I * y_max
(C) Z = A / I
(D) Z = k^2 / y_max

39. In structural beam design, a cross-section with a larger Section Modulus Z has:

(A) Greater flexural bending strength
(B) Smaller load carrying capacity
(C) Larger shear deformation
(D) Higher susceptibility to buckling

40. A structural steel I-beam is highly efficient in resisting bending moments about its major axis because:

(A) Most of its cross-sectional area is situated far from the neutral axis in flanges
(B) Its web has zero thickness
(C) It has zero product of inertia
(D) It has circular symmetry

41. A rectangle has base b = 100 mm and height h = 200 mm. Its centroidal moment of inertia I_xx is:

(A) 66.67 * 10^6 mm^4
(B) 16.67 * 10^6 mm^4
(C) 266.67 * 10^6 mm^4
(D) 33.33 * 10^6 mm^4

42. In question 41, the moment of inertia of the rectangle about its base edge is:

(A) 266.67 * 10^6 mm^4
(B) 66.67 * 10^6 mm^4
(C) 133.33 * 10^6 mm^4
(D) 533.33 * 10^6 mm^4

43. In question 41, what is the radius of gyration k_xx about the centroidal x-axis?

(A) 57.74 mm
(B) 50.00 mm
(C) 100.00 mm
(D) 28.87 mm

44. A solid circular steel shaft has diameter d = 100 mm. Its centroidal moment of inertia I_xx is:

(A) 4.91 * 10^6 mm^4
(B) 9.82 * 10^6 mm^4
(C) 1.96 * 10^6 mm^4
(D) 7.85 * 10^6 mm^4

45. In question 44, what is the polar moment of inertia J of the 100 mm diameter shaft?

(A) 9.82 * 10^6 mm^4
(B) 4.91 * 10^6 mm^4
(C) 19.63 * 10^6 mm^4
(D) 2.45 * 10^6 mm^4

46. A hollow circular column has outer diameter D = 200 mm and inner diameter d = 160 mm. Its I_xx is:

(A) 46.37 * 10^6 mm^4
(B) 78.54 * 10^6 mm^4
(C) 32.17 * 10^6 mm^4
(D) 92.74 * 10^6 mm^4

47. A structural T-section has flange 150x20 mm and web 20x130 mm. Total height is 150 mm. The centroid height y_bar from base is:

(A) 105.0 mm
(B) 75.0 mm
(C) 90.0 mm
(D) 120.0 mm

48. A symmetrical I-section has flanges 200x20 mm and web 15x260 mm (overall depth 300 mm). Its centroid lies at:

(A) 150 mm from bottom (geometric center)
(B) 120 mm from bottom
(C) 180 mm from bottom
(D) 100 mm from bottom

49. For the symmetric I-section in question 48, the total cross-sectional area is:

(A) 11,900 mm^2
(B) 8,000 mm^2
(C) 15,000 mm^2
(D) 9,500 mm^2

50. For the symmetric I-section in question 48, the centroidal moment of inertia I_xx is:

(A) 177.3 * 10^6 mm^4
(B) 125.0 * 10^6 mm^4
(C) 210.5 * 10^6 mm^4
(D) 145.2 * 10^6 mm^4

51. For the symmetric I-section in question 48, the centroidal moment of inertia I_yy (about minor axis) is:

(A) 26.74 * 10^6 mm^4
(B) 15.20 * 10^6 mm^4
(C) 45.00 * 10^6 mm^4
(D) 8.50 * 10^6 mm^4

52. A triangle has base b = 120 mm and height h = 90 mm. Its centroidal moment of inertia I_xx is:

(A) 2.43 * 10^6 mm^4
(B) 7.29 * 10^6 mm^4
(C) 4.86 * 10^6 mm^4
(D) 1.22 * 10^6 mm^4

53. For the triangle in question 52, the moment of inertia about its base edge is:

(A) 7.29 * 10^6 mm^4
(B) 2.43 * 10^6 mm^4
(C) 14.58 * 10^6 mm^4
(D) 3.65 * 10^6 mm^4

54. A circle of radius r = 40 mm is revolved about a coplanar axis at distance R = 150 mm from its center. Surface area of torus is:

(A) 236,870 mm^2
(B) 118,435 mm^2
(C) 473,741 mm^2
(D) 75,398 mm^2

55. For the torus in question 54, the total volume enclosed is:

(A) 4.74 * 10^6 mm^3
(B) 2.37 * 10^6 mm^3
(C) 9.48 * 10^6 mm^3
(D) 1.18 * 10^6 mm^3

56. A right-angled triangle with base 60 mm and height 80 mm is revolved about its 80 mm vertical leg. The volume of cone formed is:

(A) 301,593 mm^3
(B) 603,186 mm^3
(C) 150,796 mm^3
(D) 904,779 mm^3

57. A semicircular area of radius R = 60 mm has its centroid located at what distance from the straight diametral boundary?

(A) 25.46 mm
(B) 30.00 mm
(C) 20.00 mm
(D) 38.20 mm

58. An equal angle section ISA 100x100x10 mm has legs of 100 mm and thickness 10 mm. The centroid coordinates (x_bar, y_bar) from outer corner are:

(A) (28.68 mm, 28.68 mm)
(B) (35.00 mm, 35.00 mm)
(C) (25.00 mm, 25.00 mm)
(D) (50.00 mm, 50.00 mm)

59. A square plate of side 200 mm has a concentric circular hole of diameter 100 mm removed. The remaining area is:

(A) 32,146 mm^2
(B) 40,000 mm^2
(C) 7,854 mm^2
(D) 25,000 mm^2

60. For the square plate with circular hole in question 59, the centroidal moment of inertia I_xx is:

(A) 128.42 * 10^6 mm^4
(B) 133.33 * 10^6 mm^4
(C) 4.91 * 10^6 mm^4
(D) 100.00 * 10^6 mm^4

61. A composite section consists of a 100x10 mm steel strip welded to the top of a 100x100x10 mm square tube. Parallel axis theorem is used to:

(A) Transfer individual component MOIs to the combined centroidal axis
(B) Find the weight of the weld
(C) Determine Poisson's ratio
(D) Calculate plastic hinge capacity

62. The radius of gyration of a circular cross-section of radius R about any centroidal diameter is:

(A) R / 2
(B) R / 4
(C) R / sqrt(2)
(D) R * sqrt(2)

63. The radius of gyration of a rectangle of height h about its centroidal axis parallel to base is:

(A) h / sqrt(12) = 0.2887 * h
(B) h / 2
(C) h / 3
(D) h / sqrt(3)

64. If I_xx = 80 * 10^6 mm^4 and I_yy = 20 * 10^6 mm^4 for a cross-section, its polar moment of inertia J is:

(A) 100 * 10^6 mm^4
(B) 60 * 10^6 mm^4
(C) 40 * 10^6 mm^4
(D) 1600 * 10^6 mm^4

65. A channel section (C-shape) has a vertical web and two horizontal flanges. Its shear center:

(A) Lies outside the web on the horizontal symmetry axis
(B) Coincides with the centroid
(C) Lies at the mid-depth of the web
(D) Is undefined for open sections

66. When an area has two axes of symmetry, its principal axes of inertia are:

(A) The two axes of symmetry themselves
(B) Inclined at 45 degrees to the axes of symmetry
(C) Undefined
(D) Perpendicular to the plane

67. For an unequal angle section with I_xx = 150 cm^4, I_yy = 80 cm^4, and I_xy = 60 cm^4, the principal angle 2*theta_p is:

(A) -59.74 degrees
(B) +30.25 degrees
(C) -45.00 degrees
(D) 0 degrees

68. In question 67, the maximum principal moment of inertia I_max is:

(A) 185.3 cm^4
(B) 150.0 cm^4
(C) 210.0 cm^4
(D) 165.5 cm^4

69. In question 67, the minimum principal moment of inertia I_min is:

(A) 44.7 cm^4
(B) 80.0 cm^4
(C) 60.0 cm^4
(D) 25.2 cm^4

70. Mohr's circle of inertia is a graphical construction where the coordinates of points on the circle represent:

(A) (I_u, I_uv) = (Moment of Inertia, Product of Inertia)
(B) (Normal Stress, Shear Stress)
(C) (Strain, Curvature)
(D) (Area, Centroid)

71. The radius R_mohr of Mohr's circle of inertia is calculated as:

(A) sqrt(((I_xx - I_yy)/2)^2 + I_xy^2)
(B) (I_xx + I_yy) / 2
(C) sqrt(I_xx^2 + I_yy^2)
(D) I_xx - I_yy

72. The center of Mohr's circle of inertia lies on the horizontal axis at coordinate:

(A) ((I_xx + I_yy) / 2 , 0)
(B) (I_xx - I_yy , 0)
(C) (0, I_xy)
(D) (J, 0)

73. Section modulus of a solid rectangular beam of width b and depth d is:

(A) b * d^2 / 6
(B) b * d^3 / 12
(C) b^2 * d / 6
(D) b * d / 2

74. Section modulus of a solid circular cross-section of diameter d is:

(A) pi * d^3 / 32
(B) pi * d^4 / 64
(C) pi * d^3 / 16
(D) pi * d^2 / 4

75. Comparing a solid circular beam of diameter d and a square beam of side d, which has greater flexural section modulus?

(A) The square beam (Z_sq = d^3/6 = 0.167*d^3 vs Z_cir = pi*d^3/32 = 0.098*d^3)
(B) The circular beam
(C) Both have identical section moduli
(D) Depends on beam span length

76. A built-up structural steel plate girder has web plate 1200x12 mm and two flange plates 400x25 mm. Overall depth is 1250 mm. Its I_xx is:

(A) 3.91 * 10^9 mm^4
(B) 2.15 * 10^9 mm^4
(C) 5.40 * 10^9 mm^4
(D) 1.72 * 10^9 mm^4

77. In question 76, what is the elastic section modulus Z_xx of the built-up plate girder?

(A) 6.26 * 10^6 mm^3
(B) 3.13 * 10^6 mm^3
(C) 8.50 * 10^6 mm^3
(D) 4.80 * 10^6 mm^3

78. A concrete trapezoidal dam section has vertical upstream face, top crest 3 m, bottom base 9 m, and height 18 m. Centroid from upstream face is:

(A) 3.50 m
(B) 4.50 m
(C) 5.25 m
(D) 2.75 m

79. In question 78, what is the height of the centroid of the trapezoidal dam above its base?

(A) 7.00 m
(B) 6.00 m
(C) 9.00 m
(D) 8.50 m

80. A structural steel unequal angle ISA 150x100x10 mm has area 2400 mm^2. The centroid coordinates from outer corner are:

(A) (28.1 mm, 53.1 mm)
(B) (35.0 mm, 60.0 mm)
(C) (25.0 mm, 50.0 mm)
(D) (30.0 mm, 55.0 mm)

81. A column cross-section is fabricated from two ISMC 300 channels placed back-to-back with a gap s. The spacing s to make I_xx = I_yy is:

(A) Calculated to equalize major and minor axis buckling capacities
(B) Always zero
(C) Equal to the flange width
(D) s = 500 mm always

82. A parabolic spandrel area bounded by y = (h/b^2)*x^2, x = b, and y = 0 has centroid coordinates (x_bar, y_bar) equal to:

(A) (3b/4, 3h/10)
(B) (2b/3, h/3)
(C) (b/2, h/2)
(D) (4b/5, 2h/5)

83. The total area of the parabolic spandrel in question 82 is:

(A) b * h / 3
(B) 2 * b * h / 3
(C) b * h / 2
(D) b * h / 4

84. The centroidal moment of inertia I_xx of a parabolic spandrel about its base x-axis is:

(A) b * h^3 / 21
(B) b * h^3 / 12
(C) b * h^3 / 35
(D) b * h^3 / 7

85. A circular sector of radius R and included angle 2*alpha has its centroid located on the axis of symmetry at distance:

(A) (2 * R * sin(alpha)) / (3 * alpha)
(B) (4 * R * sin(alpha)) / (3 * pi)
(C) (R * sin(alpha)) / alpha
(D) R / 2

86. When alpha approaches 90 degrees (pi/2 rad), the circular sector centroid formula (2*R*sin(alpha))/(3*alpha) reduces to:

(A) 4 * R / (3 * pi) (Semicircular area centroid)
(B) 2 * R / pi
(C) R / 2
(D) 3 * R / 4

87. A thin hemispherical shell of radius R and uniform thickness has its centre of gravity located on the axis of symmetry at:

(A) R / 2 from base plane
(B) 3 * R / 8 from base plane
(C) 4 * R / (3 * pi)
(D) 2 * R / 3

88. A solid uniform hemisphere of radius R has its centre of gravity located on the axis of symmetry at:

(A) 3 * R / 8 from base plane
(B) R / 2 from base plane
(C) 4 * R / (3 * pi)
(D) 3 * R / 5

89. A solid right circular cone of height h and base radius R has its centre of gravity at what height from its base?

(A) h / 4
(B) h / 3
(C) 3 * h / 8
(D) h / 2

90. A hollow thin-walled right circular cone of height h (without base) has its centre of gravity at what height from base?

(A) h / 3
(B) h / 4
(C) h / 2
(D) 2 * h / 5

91. For a structural Z-section with equal flanges, the product of inertia I_xy about its centroidal axes is:

(A) Non-zero and negative (or positive depending on quadrant orientation)
(B) Strictly zero
(C) Equal to I_xx
(D) Undefined

92. An unequal angle section has I_xx = 250 cm^4, I_yy = 120 cm^4, and I_xy = -90 cm^4. The orientation of the major principal axis is:

(A) -27.1 degrees
(B) +45.0 degrees
(C) +15.5 degrees
(D) -60.0 degrees

93. For the angle section in question 92, the maximum principal moment of inertia I_u is:

(A) 296.3 cm^4
(B) 250.0 cm^4
(C) 350.0 cm^4
(D) 210.5 cm^4

94. For the angle section in question 92, the minimum principal moment of inertia I_v is:

(A) 73.7 cm^4
(B) 120.0 cm^4
(C) 90.0 cm^4
(D) 45.0 cm^4

95. A composite reinforced concrete column has a square cross-section 400x400 mm reinforced with four 25 mm diameter steel bars at corners (cover 40 mm). Equivalent transformed section MOI is used to:

(A) Account for modular ratio m = E_steel / E_concrete in flexural analysis
(B) Determine cement hydration rate
(C) Calculate plastic creep only
(D) Eliminate shear reinforcement

96. The polar radius of gyration k_z of a thin hollow thin-walled cylinder of mean radius R is:

(A) R
(B) R / sqrt(2)
(C) R / 2
(D) R * sqrt(2)

97. The polar radius of gyration k_z of a solid circular cylinder of radius R is:

(A) R / sqrt(2) = 0.7071 * R
(B) R
(C) R / 2
(D) 0.5 * R

98. A structural T-beam with flange 200x20 mm and web 20x180 mm has centroidal height y_bar = 142.63 mm from bottom. Its centroidal I_xx is:

(A) 27.68 * 10^6 mm^4
(B) 15.40 * 10^6 mm^4
(C) 35.20 * 10^6 mm^4
(D) 19.85 * 10^6 mm^4

99. For the T-beam in question 98, the distance to the extreme top fiber is y_top = 200 - 142.63 = 57.37 mm. The section modulus Z_top is:

(A) 482,482 mm^3
(B) 194,068 mm^3
(C) 350,000 mm^3
(D) 620,000 mm^3

100. For the T-beam in question 98, the distance to the extreme bottom fiber is y_bot = 142.63 mm. The section modulus Z_bottom is:

(A) 194,068 mm^3
(B) 482,482 mm^3
(C) 250,000 mm^3
(D) 120,000 mm^3

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Distinguish between Centre of Gravity, Centre of Mass, and Centroid of an Area. State the mathematical definitions for each and specify under what physical conditions they coincide.

Q2. State and prove both Theorems of Pappus-Guldinus for the determination of surface areas and volumes of bodies of revolution.

Q3. State and prove the Parallel Axis Theorem (Steiner's Theorem) for area moments of inertia and explain why the reference axis must be centroidal.

Q4. State and derive the Perpendicular Axis Theorem for a plane lamina and discuss its validity and limitations in engineering mechanics.

Q5. A structural steel T-section has a top flange of 200 mm x 20 mm and a vertical web of 20 mm x 180 mm (total depth 200 mm). Determine: (1) Position of centroid from bottom base, (2) Centroidal moments of inertia I_xx and I_yy, (3) Radii of gyration k_xx and k_yy.

Q6. A symmetrical I-beam has top and bottom flanges of 200 mm x 20 mm and a vertical web of 10 mm x 260 mm (overall depth 300 mm). Calculate: (1) Centroidal I_xx, (2) Centroidal I_yy, (3) Section modulus Z_xx and Z_yy.

Q7. Using the Theorems of Pappus-Guldinus, derive the exact expressions for: (1) Total surface area and volume of a circular torus, (2) Surface area and volume of a right circular cone.

Q8. Derive from first principles by direct integration the centroid and centroidal moment of inertia of a triangular area of base b and height h.

Q9. Define the Product of Inertia I_xy of a plane area, derive its parallel axis transformation formula, and prove that I_xy vanishes if the area has an axis of symmetry.

Q10. Explain the concept of Principal Moments of Inertia and Principal Axes. Derive the mathematical formulas for the principal angle theta_p and principal moments I_max and I_min.

Q11. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 11.

Q12. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 12.

Q13. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 13.

Q14. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 14.

Q15. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 15.

Q16. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 16.

Q17. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 17.

Q18. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 18.

Q19. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 19.

Q20. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 20.

Q21. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 21.

Q22. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 22.

Q23. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 23.

Q24. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 24.

Q25. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 25.

Q26. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 26.

Q27. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 27.

Q28. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 28.

Q29. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 29.

Q30. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 30.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

Proof:
Let the axis of revolution be the x-axis. Consider a differential area element dA at distance y.
Revolving dA through 2*pi radians sweeps out a circular ring of volume: dV = 2 * pi * y * dA.
Integrating over the total plane area A:
Volume = Integral(dV) = Integral(2 * pi * y * dA) = 2 * pi * Integral(y dA)
By definition of the centroid of a plane area: y_bar = Integral(y dA) / A => Integral(y dA) = y_bar * A.
Substituting yields: Volume = 2 * pi * y_bar * A. (Q.E.D.)

3. Critical Constraint:
The axis of revolution must lie in the same plane as the generating figure and must NOT intersect the generating curve or area (it may touch the boundary tangentially).

>>Solution to Question 3: State and prove the Parallel Axis Theorem (Steiner's Theorem) for area moments of inertia and explain why the reference axis must be

d2 = |y2 - y_bar| = |90.0 - 142.63| = 52.63 mm.
- For Flange 1:
I_G1 = (200 * 20^3) / 12 = 133,333 mm^4.
I_xx1 = I_G1 + A1 * d1^2 = 133,333 + 4000 * (47.37)^2 = 133,333 + 8,975,668 = 9,109,001 mm^4.
- For Web 2:
I_G2 = (20 * 180^3) / 12 = 9,720,000 mm^4.
I_xx2 = I_G2 + A2 * d2^2 = 9,720,000 + 3600 * (52.63)^2 = 9,720,000 + 9,971,687 = 19,691,687 mm^4.
Total I_xx = I_xx1 + I_xx2 = 9,109,001 + 19,691,687 = 28.80 * 10^6 mm^4.

4. Centroidal Moment of Inertia I_yy (About vertical symmetry axis):
Since both element centroids lie directly on the y-axis (d_x = 0):
I_yy = (20 * 200^3) / 12 + (180 * 20^3) / 12 = 13,333,333 + 120,000 = 13.45 * 10^6 mm^4.

5. Radii of Gyration:
k_xx = sqrt(I_xx / A) = sqrt(28.80 * 10^6 / 7600) = 61.56 mm.
k_yy = sqrt(I_yy / A) = sqrt(13.45 * 10^6 / 7600) = 42.07 mm.

>> Solution to Question 6:

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y_bar = Q_x / A = (b * h^2 / 6) / (b * h / 2) = h / 3.
Thus, the centroid is located at exactly h/3 from the base (or 2h/3 from the apex).

3. Moment of Inertia about Base Axis (I_base):
I_base = Integral_0^h y^2 dA = Integral_0^h y^2 * b * (1 - y/h) dy = b * Integral_0^h (y^2 - y^3 / h) dy
I_base = b * [y^3 / 3 - y^4 / 4h]_0^h = b * [h^3 / 3 - h^3 / 4] = b * h^3 / 12.

4. Centroidal Moment of Inertia I_xx via Parallel Axis Theorem:
Distance from centroid to base: d = h / 3.
I_base = I_centroidal + A * d^2
I_centroidal = I_base - A * d^2 = (b * h^3 / 12) - [(b * h / 2) * (h / 3)^2]
I_centroidal = (b * h^3 / 12) - (b * h^3 / 18) = b * h^3 * [(3 - 2) / 36] = b * h^3 / 36.
(Q.E.D.)

>> Solution to Question 10: Define the Product of Inertia I_xy of a plane area, derive its parallel axis transformation formula, and prove that I_xy vanishes if the area

[Problem Statement]: Explain the concept of Principal Moments of Inertia and Principal Axes. Derive the mathematical formulas for the principal a
[Detailed Step-by-Step Answer]:
1. Coordinate Rotation and Transformation of Inertia:
Let x and y be coordinate axes with known properties I_xx, I_yy, and I_xy.
Consider rotated axes (u, v) inclined at counterclockwise angle theta to (x, y).
The transformation equations for coordinates are: u = x*cos(theta) + y*sin(theta), v =
-x*sin(theta) + y*cos(theta).
Evaluating I_uu = Integral(v^2 dA) and I_uv = Integral(u*v dA) yields:
I_uu = [(I_xx + I_yy)/2] + [(I_xx - I_yy)/2] * cos(2*theta) - I_xy * sin(2*theta)
I_uv = [(I_xx - I_yy)/2] * sin(2*theta) + I_xy * cos(2*theta)

2. Derivation of Principal Angle theta_p:
Principal axes are defined as axes about which the product of inertia vanishes: I_uv = 0.
Setting I_uv = 0:
[(I_xx - I_yy)/2] * sin(2*theta_p) + I_xy * cos(2*theta_p) = 0
sin(2*theta_p) / cos(2*theta_p) = - 2 * I_xy / (I_xx - I_yy)
tan(2*theta_p) = - 2 * I_xy / (I_xx - I_yy)
This equation yields two orthogonal angles (separated by 90 degrees), defining the principal
axes.

3. Expressions for Principal Moments of Inertia (I_max, I_min):
Substituting the trigonometric values of cos(2*theta_p) and sin(2*theta_p) into I_uu:
I_max,min = (I_xx + I_yy) / 2 +/- sqrt([(I_xx - I_yy) / 2]^2 + I_xy^2)

4. Invariant Properties of Inertia:
(a) First Invariant: I_uu + I_vv = I_xx + I_yy = Polar Moment of Inertia J (Constant).
(b) Second Invariant: I_uu * I_vv - I_uv^2 = I_xx * I_yy - I_xy^2.

>> Solution for Question 11: Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 11.

>>Solution Statement: Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 14.

2. Centroidal Coordinate Calculations (x_bar, y_bar):
Subdivide into standardized sub-areas: A1, A2, A3 with local centroids (x1, y1), (x2, y2).
Total Area A = Sigma(Ai).
First moment of area: Q_y = Sigma(Ai * xi), Q_x = Sigma(Ai * yi).
Centroidal coordinates: x_bar = Q_y / A, y_bar = Q_x / A.

3. Application of Parallel Axis Theorem for Moments of Inertia:
For each element: I_xxi = I_Gi + Ai * (yi - y_bar)^2; I_yyi = I_Gi + Ai * (xi - x_bar)^2.
Summation yields combined section properties: I_xx = Sigma(I_xxi), I_yy = Sigma(I_yyi).
Polar moment of inertia: J = I_xx + I_yy; Radii of gyration: k_xx = sqrt(I_xx / A), k_yy = sqrt(I_yy / A).

4. Flexural Section Modulus & Design Implications:
Section modulus Z = I / y_max governs bending stress sigma = M / Z under civil design codes.

>>Problem Statement: Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 16.

3. Application of Parallel Axis Theorem for Moments of Inertia:
For each element: I_xxi = I_Gi + Ai * (yi - y_bar)^2; I_yyi = I_Gi + Ai * (xi - x_bar)^2.
Summation yields combined section properties: I_xx = Sigma(I_xxi), I_yy = Sigma(I_yyi).
Polar moment of inertia: J = I_xx + I_yy; Radii of gyration: k_xx = sqrt(I_xx / A), k_yy = sqrt(I_yy / A).

4. Flexural Section Modulus & Design Implications:
Section modulus Z = I / y_max governs bending stress sigma = M / Z under civil design codes.

>> Solution Statement: Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 18.

4. Flexural Section Modulus & Design Implications:
Section modulus Z = I / y_max governs bending stress sigma = M / Z under civil design codes.

>>Solution Statement: 29. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 29.

>>Solution Statement for 21. Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 21.

2. Centroidal Coordinate Calculations (x_bar, y_bar):
Subdivide into standardized sub-areas: A1, A2, A3 with local centroids (x1, y1), (x2, y2).
Total Area A = Sigma(Ai).
First moment of area: Q_y = Sigma(Ai * xi), Q_x = Sigma(Ai * yi).
Centroidal coordinates: x_bar = Q_y / A, y_bar = Q_x / A.

3. Application of Parallel Axis Theorem for Moments of Inertia:
For each element: I_xxi = I_Gi + Ai * (yi - y_bar)^2; I_yyi = I_Gi + Ai * (xi - x_bar)^2.
Summation yields combined section properties: I_xx = Sigma(I_xxi), I_yy = Sigma(I_yyi).
Polar moment of inertia: J = I_xx + I_yy; Radii of gyration: k_xx = sqrt(I_xx / A), k_yy = sqrt(I_yy / A).

4. Flexural Section Modulus & Design Implications:
Section modulus Z = I / y_max governs bending stress sigma = M / Z under civil design codes.

>> Solution to Question 25: Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 25.

3. Application of Parallel Axis Theorem for Moments of Inertia:
For each element: I_xxi = I_Gi + Ai * (yi - y_bar)^2; I_yyi = I_Gi + Ai * (xi - x_bar)^2.
Summation yields combined section properties: I_xx = Sigma(I_xxi), I_yy = Sigma(I_yyi).
Polar moment of inertia: J = I_xx + I_yy; Radii of gyration: k_xx = sqrt(I_xx / A), k_yy = sqrt(I_yy / A).

4. Flexural Section Modulus & Design Implications:
Section modulus Z = I / y_max governs bending stress sigma = M / Z under civil design codes.

>> Solution to Question 25: Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 25.

4. Flexural Section Modulus & Design Implications:
Section modulus Z = I / y_max governs bending stress sigma = M / Z under civil design codes.

>>Solution Statement: Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 29.

>>**Solution to Question 30:** Analyze the cross-sectional geometry, centroid, and moment of inertia for structural engineering problem 30.